

CAN WE KNOW ANYTHING FOR SURE?

You can be sure about things you have experienced for yourself. For example, you know that water is wet and that fire burns. You know what you had for breakfast (things in the past) and what you are wearing (things in the present), though you cannot be sure about the future. You also “know” far more things than your own experience tells you—information you get from other people, such as parents and teachers. But can you be sure it's true?

SURE THING

One thing is guaranteed, and we all have to face it—one day we are going to die. Actually, according to American statesman and scientist Benjamin Franklin (1706–1790), there are only two guarantees—death and taxes.

COMMON KNOWLEDGE

We possess a vast database of information that has been built up over generations of human existence. It is passed on in the stories parents tell their children, the books authors write, and the records people leave behind them. More and more of this information is being stored on the Internet. It is always growing and changing as people make new discoveries that alter our ideas about the past or reveal new facts about the world and the universe. Today, we know for a certainty that Earth and the other planets orbit the Sun, but for thousands of years, people believed the exact opposite—that the Sun circles around Earth. After all, both Greek philosopher Aristotle (384–322 BCE), the greatest authority of the ancient world, and the Bible, the Christian holy book, said so, and they couldn't be wrong, could they?

BACK TO SCHOOL

Well, yes they could. About 400 years ago, astronomer proved beyond a doubt that Earth goes around the Sun. They were persecuted because mathematical doubt that Earth's teachings and upset “accepted” knowledge. They opposed the Christian Church's foundations of modern science. But they paved the way for a series of revolutionary discoveries of modern astronomy, physics, and medicine—that laid the foundations of modern science. At one time doctors believed that the human body was made up of four humors (black bile, yellow bile, phlegm, and blood). That was another idea the scientific revolution helped lay to rest. Phew—some of those old cures were scary! For example, when people turned red in the sun, it was believed that they had too much blood. Bloodletting (when you purposely cut yourself) was practiced to balance their humors.

KNOWING AND BELIEVING

There is a difference between what we know for a fact to be true and what we believe to be true. Factual knowledge is based on eyewitness evidence (either our own or that of other people), rational explanation, and scientific proof. Belief is based on the hope or confidence that something is true. People of all religious faiths have no doubt that their beliefs are the truth, but they cannot prove for certain that God or the supernatural exists.

“KNOWLEDGE IS AN UNENDING ADVENTURE AT THE EDGE OF UNCERTAINTY.”

Jacob Bronowski (1908–1974),
British scientist

